number of up-and-down cycles of oscillation that a wave completes every second. It turns out that the frequency is determined by the wavelength and vice versa: longer wavelengths imply lower frequency; shorter wavelengths imply higher frequency. To see why, think of what happens when you produce waves by shaking a long rope that is tied down at one end. To generate a long wavelength, you leisurely shake your end up and down. The frequency of the waves matches the number of cycles per second your arm goes through and is consequently fairly low. But to generate short wavelengths you shake your end more frantically—more frequently, so to speak—and this yields a higher-frequency wave. Finally, physicists use the term amplitude to describe the maximum height or depth of a wave, as also illustrated in Figure 4.2.

In case you find electromagnetic waves a bit abstract, another good analogy to keep in mind are the waves that are produced by plucking a violin string. Different wave frequencies correspond to different musical notes: the higher the frequency, the higher the note. The amplitude of a wave on a violin string is determined by how hard you pluck it. A harder pluck means that you put more energy into the wave disturbance; more energy therefore corresponds to a larger amplitude. You can hear this, as the resulting tone is louder. Similarly, less energy corresponds to a smaller amplitude and a lower volume of sound.

By making use of nineteenth-century thermodynamics, physicists were able to determine how much energy the hot walls of the oven would pump into electromagnetic waves of each allowed wavelength-how hard the walls would, in effect, “pluck” each wave. The result they found is simple to state: Each of the allowed waves—regardless of its wavelength—carries the same amount of energy (with the precise amount determined by the temperature of the oven). In other words, all of the possible wave patterns within the oven are on completely equal footing when it comes to the amount of energy they embody.

At first this seems like an interesting, albeit innocuous, result. It isn’t. It spells the downfall of what has come to be known as classical physics. The reason is this: Even though requiring that all waves have a whole number of peaks and troughs rules out an enormous variety of conceivable wave patterns in the oven, there are still an infinite number that are possible—those with ever more peaks and troughs.